

Portfolio

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PPG Team Final Design



PROBE Lab original design

The Problem

The PROBE Lab at Brown University developed a novel PPG sensor designed to maintain signal accuracy across varying melanin levels for outpatient monitoring applications.

However, the existing wearable form factor was:

Bulky
Uncomfortable
Difficult to use



PPG Team Design

This project focused on designing a more compact, wearable, and ergonomically stable encapsulation for testing on Brown students with a final design that is:

Wearable and user-friendly design

Integrated watch-style system improved comfort and usability.

Compact single-enclosure form factor

Reduced bulk and improved wrist conformity.

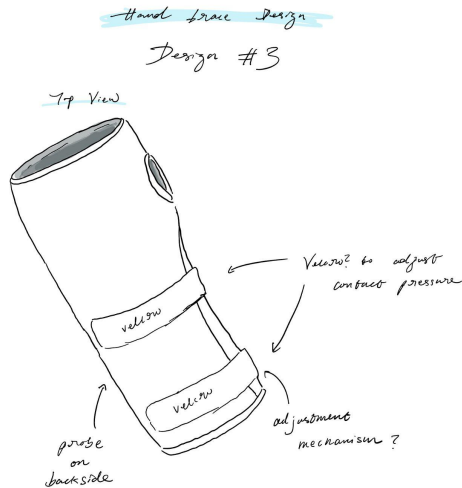
Improved signal stability

Maintained consistent sensor-to-skin contact for faster acquisition.

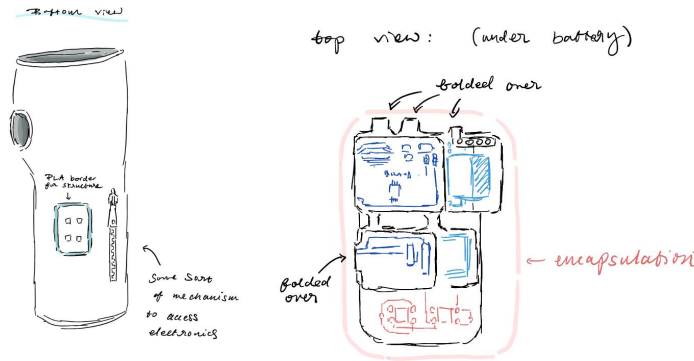
Initial Sketches

Watch & Cuff Design

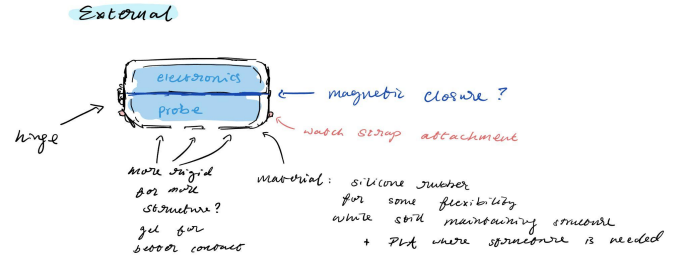
My initial sketches explored both watch-style and cuff-style encapsulations, using a folded PCB architecture to reduce device footprint. Early concepts investigated weighted internal attachments for PCB stabilization before transitioning to the support rod used in the final design.



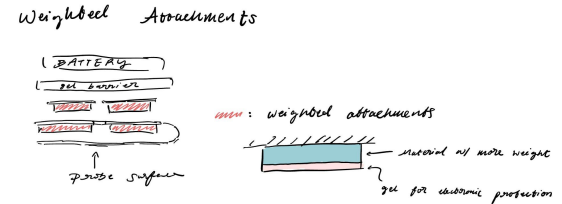
Cuff/brace design for contact pressure and stability



Folded over internal PCB layout



External encapsulation



Internal weighted attachments for PCB separation and protection

Working Prototypes - Iterations

Prototype 1 - low fidelity prototype

Cardboard wearables conceptualized based on initial sketches exploring watch and cuff-style form factors. Early prototypes focused on placement, comfort, and attachment mechanisms. Watch-style prototypes created by Anika.

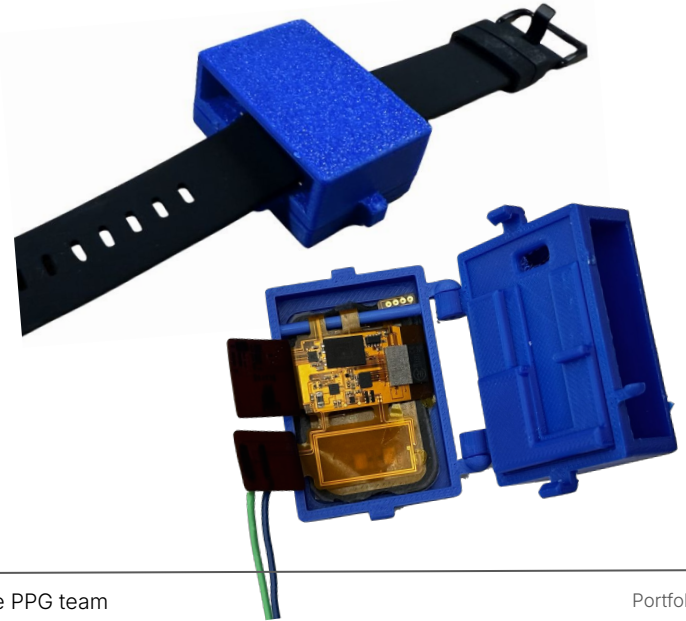


Prototype 2 - resin printed

Selected the watch-style direction as a team and developed a dual-encapsulation design for unfolded PCB "ears" using CAD and resin printing. The design introduced excessive bulk, imprecise fabrication, and increased risk of PCB stress between the enclosures.

Prototype 3 - 3D Printed

Revised into single enclosure with folded PCB integration, internal extrusions, and a battery compartment. Incorrect battery measurements and unnecessary material usage meant further bulk improvement for the final prototype.

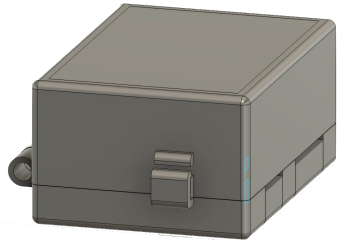
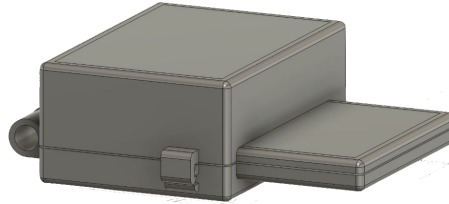
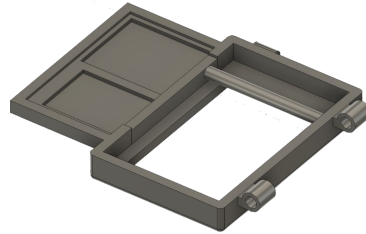
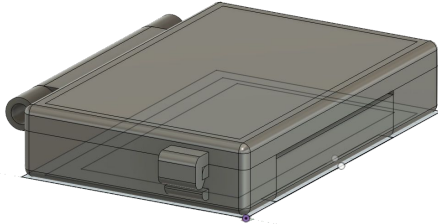


CAD Models - Iterations

I designed CAD iterations exploring enclosure geometry, PCB integration, and wearable form factor. Additional minor refinements and dimensional adjustments occurred between the versions shown. Initial ideas for the models were discussed as a team.

Iteration 1

Separate enclosures for the sensor region and PCB "ears" based on bend radius constraints, resulting in uneven contact pressure and excessive bulk. Internal PCB support rod designed by Anika.



Iteration 3

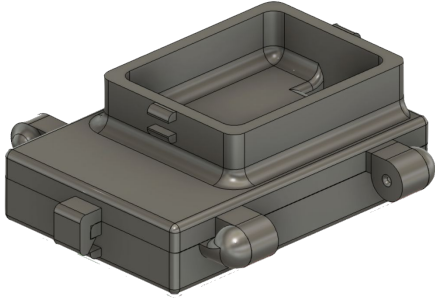
Added internal extrusions and a dedicated battery compartment to improve PCB stability and organization, with refinements focused on reducing bulk and improving wearability.



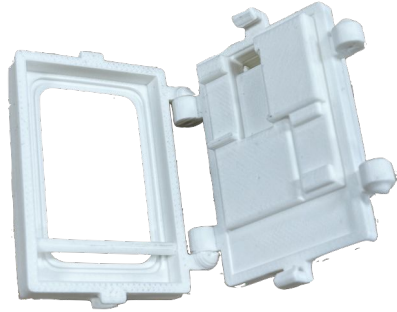
Iteration 2

Combined into a single enclosure while retaining PCB "ears" to avoid folding the flexible PCB. Later testing showed folding was feasible, making the extensions unnecessary.

CAD Modeling - Final

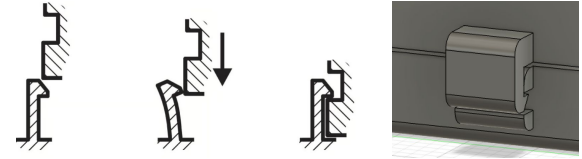


The final CAD model refined the previous iteration through PROBE Lab testing by improving wire routing, watch strap attachment, and extrusion geometry for better PCB stability and contact pressure. These changes reduced bulk, improved wrist conformity, and optimized internal organization.



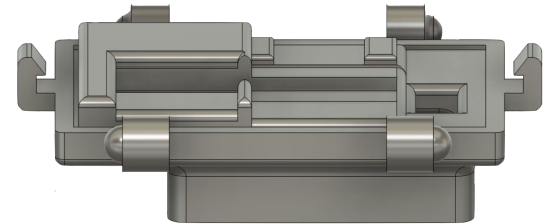
Hinge & watch strap integration

Snap-fit hinge mechanism with interlocking ball joints for secure assembly and smooth movement, with integrated openings for watch strap attachment.



Closure mechanism

Cantilever snap-fit closure integrated into the main housing for secure fastening, simplified fabrication, and improved durability.



Extrusions for contact pressure

Internal extrusions designed around PCB dimensions to maintain consistent contact pressure through the silicone layer while preserving battery wire clearance. Multiple iterations optimized fit and pressure distribution.

Engineering Testing

In collaboration with the PROBE Lab, a functioning PCB was integrated into the final encapsulation to evaluate contact pressure and PPG signal acquisition using force sensor calibration and time-to-signal testing. This resulted in:

Contact pressure measured at 43.94 mmHg, within clinically validated PPG sensing range

Signal acquisition improved from 7:28 to 4:37, producing clearer and more stable waveforms

Improved sensor-to-skin contact reduced motion artifacts and increased signal consistency

Testing identified remaining durability and battery integration limitations



Testing validated improvements in form factor, wearability, and signal acquisition performance.

Next steps: refine battery enclosure, improve durability of repeated PCB folding, and optimize long-term wearability.